OBO Foundry in 2021

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ABSTRACT

The Open Biomedical Ontologies (OBO) Foundry (http://obofoundry.org) promotes sharing of biomedical ontologies and vocabularies, supporting collaborative ontology development and utilization of ontology design principles. There are over 200 biomedical ontologies in the OBO Library. The OBO Foundry is led by a collective of ontology developers who are committed to collaboration and adherence to shared principles. Collaborative development of ontologies using best practices and principles provides multiple advantages for biocurators.

The OBO Foundry promotes consistent identifier use and metadata schemas for its resources and actively maintains a system of permanent URLs for those identifiers. This in turn enables resolution of ontology terms in a web browser for human curators, and provides a standard, computable way of referencing each term. To meet the requirements for inclusion in the OBO Foundry, ontologies must have a contact person and terms should have a clear label, definition, and source of definition (e.g. PubMed ID). The use of consistent identifiers ensures terms never disappear: even if they become obsolete, their identifiers remain valid, allowing curators to find the term corresponding to their annotation. The OBO Foundry's GitHub-backed website allows ontology editors to directly update the resource information, thus ensuring continual access to the latest information.

Developing ontologies that are easily re-usable and compatible with each other is a difficult process. The OBO Foundry principles facilitate this process by introducing shared conventions, such as standard identifiers, recommendations for open licenses, documentation, etc. In our growing community, it is hard to provide concrete development advice to implement these shared principles. This is especially true for new candidate ontologies, whose developers frequently seek advice on how to make their ontology more useful for others. To address this issue, the OBO Foundry developed a suite of tools, such as ROBOT, the Ontology Development Kit, and the OBO Dashboard, that make checking for errors (quality control) and conformance to OBO Foundry principles easier. The OBO Dashboard visually displays the results of OBO principle conformance checks and ROBOT reports, to help identify areas for improvement. The OBO Foundry actively encourages ontology developers to join the community, submit their ontologies for review, and participate in discussions on shaping the Foundry principles.

OBO Foundry and Library

Suite of integrated ontologies that are:

- Logically well formed
- Scientifically accurate
- Interoperable and interconnected
- Non-overlapping

A collective of ontology developers committed to:

- Collaboration
- Shared standards

Examples

- GO: for describing function of genes
- OBI: for describing experiments
- ENVO: environment and habitat

OBO Working groups:

- Technical
- Editorial
- Outreach

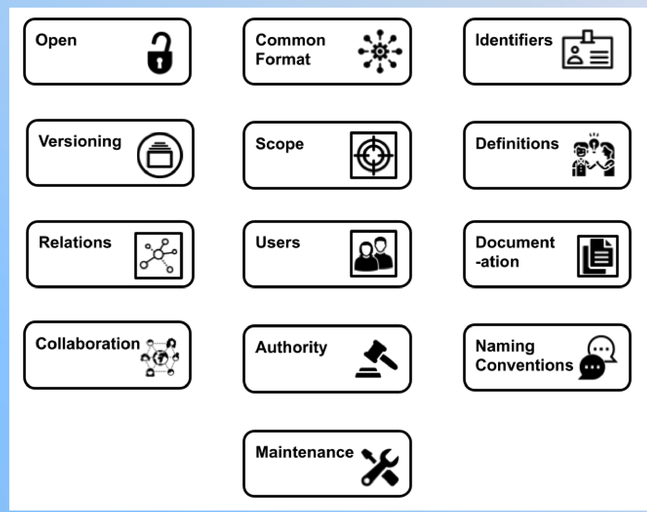


Figure 1: Illustration of the principles around which the OBO Foundry was built

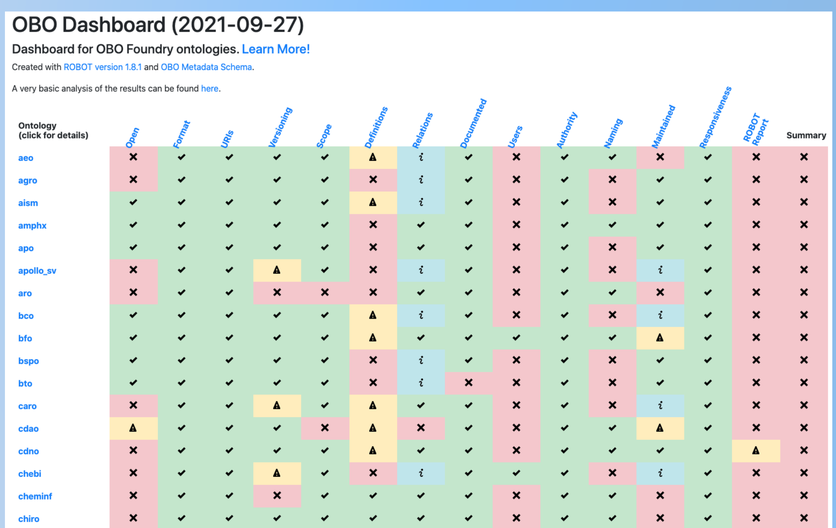


Figure 2: The OBO Dashboard (truncated). The rows represent OBO ontologies (of which the first 15 in alphabetical order are shown here) and the columns are the OBO principles. The final column (Summary) shows whether the ontology passed all of the tests. Clicking on the ontology ID in the far left column directs to a detailed report page. http://dashboard.obofoundry.org/dashboard/index.html

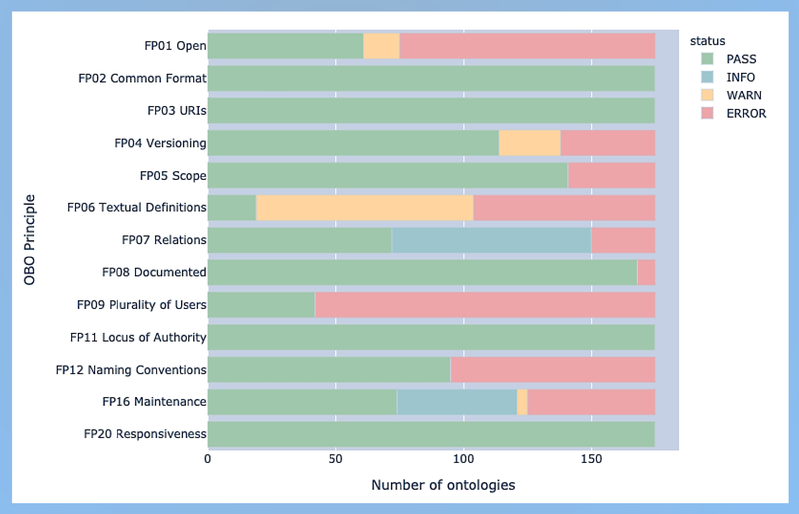


Figure 3: Summary of principle conformance across all active OBO Foundry ontologies in May 2021 http://dashboard.obofoundry.org/dashboard/analysis.html

The Ontology Development Kit

Provides a way of creating an ontology project ready for pushing to GitHub, with a number of features in place:

- A Makefile that specifies your release workflow, including building imports, creating reports and running tests
- Continuous integration: A .travis.yml file that configures Travis-CI to check any Pull Requests using ROBOT
- A standard directory layout that makes working with different projects easier and more predictable
- Standardized documentation and additional file artifacts
- A procedure for releasing your ontologies using the GitHub release mechanism
- The overall aim is to borrow as much from modern software engineering practice as possible and apply to the ontology development lifecycle.

The basic idea is fairly simple: a template folder contains a canonical repository layout, this is copied into a target area, with template variables substituted for user-supplied ones.

https://github.com/INCATools/ontology-development-kit

ROBOT

ROBOT is an OBO tool that makes working with Open Biomedical Ontologies much simpler and easier! It can be used as a command-line tool or as a library for any language on the Java Virtual Machine. ROBOT facilitates these activities:

- Manual editing
- Repetitive tasks
- Automated workflows
- Quality assurance
- Modular development
- Modular releases
- Other commands

http://robot.obolibrary.org/

PARTICIPATE

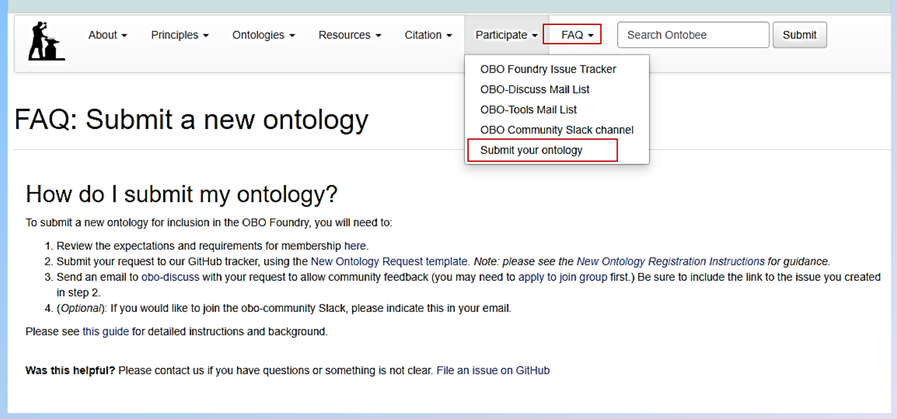
Issue Tracker: https://github.com/OBOFoundry/OBOFoundry.github.io/issues

OBO Discuss Mail List: https://groups.google.com/g/obo-discuss

OBO Tools Mail List: https://groups.google.com/g/obo-tools/members

OBO Slack Channel: https://obo-communitygroup.slack.com/

SUBMIT



http://obofoundry.org